



Application ID:	1017-2025-00465	Administering Organization:	McGill University
Applicant:	Harper, Sam	Funding Opportunity:	Research on Research Joint Initiative
Title:	The Influence of Reviewer Expertise and Engagement on Peer Review of Grants		

Application Details

Language of the application
English

Does your proposal involve Indigenous research as defined by SSHRC?
No

List of Participants

Participant	Primary Affiliation
Harper, Sam - Project Director Professor McGill University, Epidemiology, Biostatistics & Occupational Health Canada	Professor McGill University , Epidemiology, Biostatistics & Occupational Health Canada
Nandi, Arijit - Co-Applicant Associate Professor Associate Professor McGill University, Epidemiology & Equity, Ethics, and Policy Canada	Associate Professor McGill University , Epidemiology & Equity, Ethics, and Policy Canada

Partner Organizations

Contact Information

Organization Information



Fields of Research

Primary	Code	Division / Group / Class / Subclass (Field)
✓	RDF3030299	Medical, health and life sciences / Health sciences / Public and population health / Public and population health, n.e.c.
	RDF5060216	Social sciences / Political science and policy administration / Policy and policy administration / Research, science and technology policy

Keywords

Meta-science

Peer review

Research awards

Equity



Summary of Proposal

Summary of Proposal

This project investigates how the peer review process used by CIHR for Project Grant applications influences the fairness and reliability of funding decisions. Specifically, we will examine how differences in application scores emerge over the course of panel group discussion. We will compare scores given by the small number of reviewers who thoroughly read each proposal and the final scores assigned by the full committee after group discussion. We will assess whether the composition of the review panel and the characteristics of its members, particularly their self-described expertise, level of engagement with the application, and gender may affect how scores differ from the consensus. In addition, we will also focus on whether any differences by reviewer characteristics may systematically affect applicants with respect to gender and career stage.

Although panel group discussions can help clarify an application's strengths and weaknesses, there remain longstanding concerns that it may also introduce biases. When only a subset of panel members reads an application carefully, those who have not done so might rely on the opinions of their more informed colleagues or on simple heuristics, which can lead to less objective final scores. This process, influenced by group dynamics such as conformity and dominant voices, might lead to unfair funding outcomes, especially for applications from early career researchers or underrepresented groups.

To address these issues, we will analyze detailed application-level data provided by CIHR's Funding Analytics Team. This data includes information on consensus and final scores, funding status, and attributes of both applications and reviewers. Using multilevel statistical models, we will quantify how the consensus-final score difference is affected by reviewer engagement and expertise. In addition, we will evaluate alternative evaluation methods—such as weighting the scores of reviewers who read the application more carefully or introducing partial randomization for borderline cases—to determine whether these approaches might lead to more reliable and equitable funding decisions.

Expected outcomes include new evidence on whether the current peer review process optimizes fairness in judgments or disadvantages certain applicant groups. We will build skills in data science and open, reproducible research practices for trainees, and share the findings widely through publications, conferences, and policy briefs.

Proposed Budget

Year 1	Year 2
\$83,809	\$85,899
Year 3	Year 4
\$22,309	\$0
Year 5	Year 6
\$0	\$0
Year 7	Year 8
\$0	\$0
Year 9	Year 10
\$0	\$0
Total	
\$192,017	

Certifications, Licenses and Permits

Certification Requirements

Does your proposed research involve the use of human beings as research subjects?

No

Will any phase of the proposed research take place outdoors?

No

Sensitive Technology Research Areas

Will the proposed research and related activities funded by this grant aim to advance any of the listed STRAs?

No

Reviewer Exclusions

Name	Department and Organization
No information to display	



Supporting Documents

Document Name	Stage	Status
Project Description	Application	Attached
List of references or bibliography	Application	Attached
Funds requested from SSHRC	Application	Attached
Budget justification (maximum 2 pages)	Application	Attached
Funds requested from other sources	Application	Not Attached
Letters of support from partner organizations	Application	Not Attached
List of potential partner organizations	Application	Not Attached



Sam Harper

CV Documents

Document Name	Stage	Status
CV	Application	Attached

Affiliations (Employment)

Date	Organization and Department	Position
2010-08-01 - present	McGill University - Epidemiology, Biostatistics & Occupational Health	Professor



Arijit Nandi

CV Documents

Document Name	Stage	Status
CV	Application	Attached

Affiliations (Employment)

Date	Organization and Department	Position
2010-09-01 - present	McGill University - Epidemiology & Equity, Ethics, and Policy	Associate Professor



Attachments

Sam Harper - Curriculum Vitae

1 Personal Information

Current position

- Full Professor, Department of Epidemiology, Biostatistics & Occupational Health, McGill University

Education:

- PhD (*Epidemiologic Science*), University of Michigan Ann Arbor, USA 2005.
- MScPH (*Epidemiology*), University of South Carolina, USA 1999
- BA (*Biology*), Westminster College, USA, 1995

Website: samharper.org

2 Personal statement

My background is well suited to direct this project. I have a strong track record of research funding and productivity. Since 2019 I have led or co-led 5 research grants (both small and large) and generated substantial research output, with nearly 6000 citations since 2020. I have supervised and mentored over 15 trainees while also serving as Director of the PhD Program in Epidemiology at McGill from 2014-2024.

This proposal brings together two areas of research that have been passions of mine over the course of my career: equity and research transparency. Throughout my research career I have maintained a strong research interest in both the measurement and analysis of social inequalities in health, and I am eager to extend this work to assessing inequalities in how the grant peer system may (or may not) contribute to differences in funding success for vulnerable populations. I have also long maintained strong interests in metascience, reproducible research, and health equity, as noted below. Moreover, my interest in this project also derives from my personal experience. Over the past several years my service to CIHR in the capacity of Scientific Officer for the Public, Community and Population Health Committee has generated important questions for me regarding fairness in the peer review process for grants that I hope to explore in more detail with this proposal.

My work past integrating surveillance and policy on health inequalities has had an impact on policy and practice in Canada and internationally. I was a co-founding member of the McGill Public Policy and Population Health Observatory (3PO) since 2016. My international reputation as a scholar also led to an Endowed Professorship in *Impact of Health and Social Policy on Health Inequalities* at Erasmus University Medical Center, Rotterdam from 2018-2021.

3 Most significant contributions

Measuring and monitoring inequalities are crucial activities for providing evidence on the impact of interventions and policies on vulnerable populations, including for how structures such as grant peer review may disproportionately affect vulnerable subgroups. My prior work has provided key methodological insights and tools to advance the science of measuring health inequalities. Inequality is a value-laden construct, and my early work demonstrated the need to integrate ethical parameters into inequality measurement since different inequality metrics can provide diverging evidence on inequalities. This work led to two monographs solicited by the US National Cancer Institute and several highly-cited papers in top public health journals. My subsequent work in this area has been extended globally by my trainees.

My research on this topic has led to international recognition. The Pan American Health Organization (PAHO) translated one of my key papers into Spanish and my online health inequalities training [module](#)

into Spanish and Portuguese. I was chosen to serve as a member of the World Health Organization's (WHO) 2010-11 *Scientific Resource Group on Equity Analysis and Research* from over 450 nominations from around the world. I drafted the first WHO guidelines on monitoring social inequalities in health, which served as the foundation for the first WHO Handbook on Measuring Health Inequalities (Hosseinpoor *et al.*, 2013). I serve on the Steering Committee for the Public Health Agency of Canada's *Pan-Canadian Health Inequalities Reporting Initiative*. In 2022 I was invited to join the WHO's Expert Review Group for the development of the second edition of their *Health Inequality Monitoring Handbook*. I have been invited to provide technical advice and research workshops to PAHO, the US CDC, the Office of Environmental Justice at the US EPA, and the US National Cancer Institute. Most recently, I was invited by the US National Academy of Sciences, Engineering & Medicine to discuss my research on measuring inequalities for at their 2023 roundtable [series](#) on Population Health Improvement: *The Ecosystem of Health Equity Measures: A Workshop*.

Relevant recent papers:

- Chen D, Momen NC, Ejlskov L, Bødkergaard K, Werenberg Dreier J, et al. Socioeconomic inequalities in mortality associated with mental disorders: a population-based cohort study. *World Psychiatry* 2025;24(1):92-102.
- Hetherington E*, Darling E, Harper S, Nguyen F, Schummers L, et al. Inequalities in access to prenatal care during the COVID-19 pandemic: Analysis of a population-based cohort. *Paediatr Perinat Epidemiol* 2024;38:291–301.
- Carabali M, Harper S, Lima Neto AS, Dos Santos De Sousa G, Caprara A, Restrepo BN, et al. Decomposition of socioeconomic inequalities in arboviral diseases in Brazil and Colombia (2007-2017). *Trans R Soc Trop Med Hyg* 2022;116(8):717-726.
- Capurro DA*, Harper S. Socioeconomic inequalities in health care utilization in Paraguay: Description of trends from 1999 to 2018. *J Health Serv Res Policy* 2022;27(3):180-189.
- Sreeramareddy CT, Harper S. Trends in educational and wealth inequalities in adult tobacco use in Nepal 2001-2016: secondary data analyses of four Demographic and Health Surveys. *BMJ Open*. 2019 Sep 6;9(9):e029712.

Foundational papers:

- Harper S, Lynch J. Measuring and Decomposing Health Inequalities. In: Oakes JM, Kaufman JS (eds). *Methods in Social Epidemiology*, 2nd Edition; San Francisco, CA: Jossey-Bass, 2017, pp. 91-131.
- Harper S, King N, Meersman SC, Breen N, Reichman ME, Lynch J. Implicit value judgments in the measurement of health inequalities. *Milbank Quarterly* 2010;88:4-29.
- Harper S, Lynch J, Meersman SC, Breen N, Davis WW, Reichman ME. An overview of methods for monitoring social disparities in cancer with an example using trends in lung cancer incidence by socioeconomic position and race-ethnicity, 1992-2004. *Am J Epidemiol* 2008;167(8):889-99.
- Hosseinpoor AR et al. Handbook on health inequality monitoring: with a special focus on low-and middle-income countries. World Health Organization, 2013

Metascience and Reproducible Research

I also maintain a strong commitment to methodological transparency and reproducible science (Austin *et al.*, 2016; Hamra, Goldstein and Harper, 2019; Harper, 2019a; Goldstein, Hamra and Harper, 2020). When possible, I have made the raw data and statistical code to reproduce the tables and figures in my team's published papers freely available by uploading them to public repositories such as [Dataverse](#) and Open Science Foundation [page](#). I have uploaded data and code for more than a dozen of my published

papers, which have been downloaded over 12,000 times. I was invited to give a [workshop](#), *Reproducible Research: Why and How*, by the Society for Epidemiological Research, which covered scientific integrity problems, design and analytic solutions, dissemination and examples of how to conduct reproducible research. I also sought out more explicit training and took the Research Transparency and Reproducibility Training (RT2) course given by the Berkeley Insititute for Transparent Social Science in 2019. Finally, I was selected and served as a 2019-2020 Faculty Fellow at [Project TIER](#) that focuses on Teaching Integrity in Empirical Research.

Relevant papers:

- Austin N*, Harper S, Kaufman JS, Hamra G. Challenges in Reproducing Results from Publicly Available Data: An Example of Sexual Orientation and Cardiovascular Disease Risk. *J Epidemiol Community Health* 2016;70(8):807-12.
- Goldstein ND, Hamra GB, Harper S. Are Descriptions of Methods Alone Sufficient for Study Reproducibility? An Example From the Cardiovascular Literature. *Epidemiology*. 2020 Mar;31(2):184-188.
- Hamra GB, Goldstein ND, Harper S. Resource Sharing to Improve Research Quality. *Journal of the American Heart Association*. 2019;8(15):e012292.
- Harper S. A future for observational epidemiology: Clarity, credibility, transparency. *Am J Epidemiol* 2019;188:840-845.

Evaluating Interventions

This project will also provide potential insights into how changes in the peer review system (e.g., virtual vs. in-person panels) may have affected funding. I have a substantial track record, much of it with my co-applicant Arijit Nandi, in conducting studies to estimate the impact of interventions on overall outcomes and inequalities. Thematically, this work all focuses on moving beyond surveillance and associations to get credible causal estimates of social and economic intervention effects, largely using quasi-experimental and experimental methods. I have conducted extensive work on the impacts of US mandatory seat belt laws on seatbelt use (Harper *et al.*, 2014), traffic crash deaths (Harper and Strumpf, 2017; Harper, 2019b), and police discrimination (Riddell *et al.*, 2020). I have also provided strong evidence on the impacts of medical marijuana legalization (Harper, Strumpf and Kaufman, 2012; Windle *et al.*, 2022), tobacco taxes (Manivong, Harper and Strumpf, 2017), abortion restrictions (Austin and Harper, 2019; Hawkins *et al.*, 2023), clinical obstetric interventions (Hutcheon *et al.*, 2017; 2020; 2022), as well as studies looking at the impact of economic recessions on health (Harper *et al.*, 2015; Harper and Bruckner, 2017).

To expand this work in 2016 I co-founded the Public Policy and Population Health Observatory with Arijit Nandi ([3PO](#)) at McGill to promote an integrated research agenda, create and publish policy databases, and evaluate the impacts of interventions on health. We have worked closely with academic and non-academic partners to develop and answer research questions and disseminate findings, as well as training doctoral students from McGill and six other countries. A large part of this work has produced high-profile papers on the impacts of paid leave policies (Nandi *et al.*, 2016; 2018). In addition to analysis of secondary data, through 3PO I have also (with partners) developed and implemented two large-scale randomized evaluations, one on daycare in India (Nandi *et al.*, 2020) and one on a complex maternal and newborn health intervention in Tanzania (Hetherington *et al.*, 2023). Our evidence on affordable daycare impacts in India was recently [featured](#) by MIT's Poverty Action Lab.

My work on the use and value of experimental and quasi-experimental designs for public health has led to international recognition from journals through solicited commentaries on the future of epidemiology (Harper and Strumpf, 2012; Harper, 2019a). I was invited to give the keynote address at the 2019 Canadian

Society for Epidemiology & Biostatistics Annual Meeting, as well as three international talks on the value of natural experiments for public health at [UC Berkeley](#), [Drexel](#), and [Erasmus University](#).

Relevant Recent Papers:

- Nandi A, Agarwal P, Chandrashekar A, Maloney S, Richardson R, Thakur L, Harper S. Access to affordable daycare and women's mental health in Rajasthan, India: Evidence from a cluster-randomised social intervention. *J Glob Health* 2024;14:04063.
- Hetherington E*, Harper S, Davidson R, Festo C, Lampkin N, Mtenga S, Teixeira C, Vincent I, Nandi A. Impact evaluation of the TAMANI project to improve maternal and child health in Tanzania. *J Epidemiol Community Health* 2023;77(6):410-6.
- Nandi A, Charters TJ, Quamruzzaman A, Strumpf EC, Kaufman JS, Heymann J, Mukherji A, Harper S. Health care services use, stillbirth, and neonatal and infant survival following implementation of the Maternal Health Voucher Scheme in Bangladesh: A difference-in-differences analysis of Bangladesh Demographic and Health Survey data, 2000 to 2016. *PLoS Med* 2022;19(8).
- Lopes FV*, Bakx P, Harper S, Ravesteijn B, Van Ourti T. The effects of supported housing for individuals with mental disorders. *Health Econ* 2022;31(S2):115-133.
- Hawkins SS, Harper S, Baum CF, Kaufman JS. Associations between State-Level Changes in Reproductive Health Services and Indicators of Severe Maternal Morbidity. *JAMA Pediatr* 2023;177(1):93-95.
- Harper S. Would stronger seat belt laws reduce motor vehicle crash deaths? A semi-Bayesian analysis. *Epidemiology* 2019;30:380-7.
- Harper S, Palayew A. The annual cannabis holiday and fatal traffic crashes. *Inj Prev* 2019;25:433–437.

4 Other contributions to knowledge

I have a strong record of service and my expertise is routinely sought by my peers. I have conducted over 180 peer reviews across a wide range of disciplines including medicine, epidemiology, public health, biostatistics, economics, and general social science. I have served on numerous grant panels for Canada (including several years as a Scientific Officer for CIHR) and international funding agencies, in addition to routinely serving a session organizer and reviewer of abstracts for annual meetings at the leading conferences in my discipline. I have also been identified as a *Top Reviewer* by Web of Science platform and earned an award as a *Top reviewers in Social Sciences - September 2019*.

Ad hoc reviews

Epidemiology and Public Health

American Journal of Public Health; Annals of Epidemiology; Epidemiology; International Journal of Epidemiology; Journal of Epidemiology & Community Health; Social Science & Medicine; Scandinavian Journal of Public Health; BMC Public Health; Paediatrics & Child Health; International Journal for Equity in Health; Caries Research; Journal of Urban Health; Health & Place; American Journal of Epidemiology; International Journal of Obesity; Cancer Management and Research; Open Medicine; International Journal of Pediatric Obesity; Cancer Epidemiology, Biomarkers & Prevention; British Journal of Cancer; BMC Cancer; European Journal of Public Health; Cancer Causes & Control; PLOS ONE; Addiction; Population Health Metrics; Global Health Action; Traffic Injury Prevention; Cancer; Oxford University Press (book proposal); Journal of Drug Issues; Environmental Health; Nature Communications Medicine; Nature; Nature Medicine; The Lancet Regional Health – Southeast Asia

General Medicine

Annals of Internal Medicine; BMJ (top 10% of peer reviewers in 2007); The Lancet; PLOS Medicine; JAMA; New England Journal of Medicine; Canadian Medical Association Journal; JAMA Pediatrics; Canadian J Cardiology

Health Policy

Milbank Quarterly; Health Affairs; Health Policy; Health Services Research; Evaluation Review

Social Science

Demography; Population Studies; Social Compass; Social Science History; Sociological Focus; Journal of Health and Social Behavior; Population Research and Policy Review; Journal of Health and Social Behavior

Biostatistics

Journal of the Royal Statistical Society (A); Statistics in Medicine

Economics

B.E. Journal of Economics and Policy; Health Economics; Oxford Development Studies; Journal of Economic Issues; Applied Economics Letters; American Journal of Economics & Sociology

Grant Reviews

Panel Member

- | | |
|-----------|---|
| 2018-2023 | Scientific Officer, Canadian Institutes for Health Research (CIHR), Public, Community & Population Health Committee |
| 2017-2018 | Member, Canadian Institutes for Health Research (CIHR), Public, Community & Population Health Committee |

Ad hoc reviews

- | | |
|------|---|
| 2021 | Robert Wood Johnson Foundation Interdisciplinary Research Leaders Competition |
| 2019 | CIHR Canada Research Chairs (Tier 1) |
| 2019 | Faculty of Medicine, Internal grant review (2) |

Recent Invited Lectures, Talks, Presentations

In the past 6 years I have also been invited to give international lectures on the history and application of multilevel models in social epidemiology, as well as talks on the measurement of health inequalities, and quasi-experimental study designs.

2024	Keynote Address: <i>Multilevel Models in Social Epidemiology: Past, Present, and Future</i> . Danish Epidemiology Society Annual Meeting, Aarhus University, Aarhus, DK Nov 7, 2024
2024	Invited Workshop: <i>Difference-in-Differences for Social Epidemiology</i> . Danish Epidemiology Society Annual Meeting, Aarhus University, Aarhus, DK Nov 6, 2024
2024	<i>Despair and Disadvantage: Some Questions</i> Presentation given at the MORTAL workshop, Nuffield College, Oxford University, UK Jun 10-11, 2024
2023	<i>Measuring Health Equity: Beyond Description</i> . US National Academy of Sciences, Engineering, & Medicine, The Ecosystem of Health Equity Measures: A Workshop, Oakland, CA, USA June 21, 2023 (remote)
2023	<i>A Quarter Century of Multilevel Models in Social Epidemiology: A Short Story in Three Acts</i> . Society for Epidemiologic Research Annual Meeting, Portland, OR, USA June 14, 2023
2020	Invited Workshop: <i>Reproducible Research: Why and How</i> . Pre-conference Workshop for the Society for Epidemiologic Research Annual Meeting (virtual) October 30, 2020

My expertise on measuring health inequalities has also been sought by national and international organizations, where I serve as a member of several standing committees:

2024-present	Member, Expert Review Group on <i>Collaborative Health Equity Measurement</i> , Maternal and Child Health Bureau, US Health Resources and Services Administration
2023-present	Member, Expert Review Group, <i>Health Inequality Monitoring Handbook</i> , World Health Organization
2017-present	Member, Working Group, <i>Health Equity Toolkit Assessment</i> , World Health Organization
2022-present	Steering Group Member, <i>Pan-Canadian Health Inequalities Reporting Initiative</i> , Public Health Agency of Canada
2019-2021	Working Group Member, <i>Health Inequalities Reporting Initiative</i> , Canadian Institute for Health Information

5 Supervisory and mentoring activities

I have strong track record of supervision. Below I list the postdocs, doctoral students, and master's degree students I have supervised or co-supervised in the past 6 years.

Postdoctoral fellows

2020-2023	Erin Hetherington
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Doctoral students

2022-present	Siobhan Carroll (co-supervised with Jill Baumgartner)
2021-present	Wenlu Yuan (co-supervised with Jill Baumgartner)
2021-present	Peter Socha (co-supervised with Jennifer Hutcheon)
2021-present	Talia Sternbach (co-supervised with Jill Baumgartner)
2021-present	Imen Farhat (co-supervised with Dimitra Panagiotoglu)
2020-2024	Francisca Vargas Lopez (Erasmus MC)
2018-2023	Walid Al-Soneidar
2017-present	Diego Capurro Fernandez
2014-2019	Nichole Austin

Master's degree students

2021-2022	Melia Alcantara (co-supervised with Seungmi Yang)
2020-2021	Peter Socha (co-supervised with Jennifer Hutcheon)
2018-2020	Hiba El-Haj (co-supervised with Shelley Clarke)
2019-2021	Talia Sternbach (co-supervised with Jill Baumgartner)

6 Other relevant information

None.

Arijit Nandi – Curriculum Vitae

1 Personal Information

Current position

- Associate Professor; Department of Equity, Ethics, and Policy & Department of Epidemiology, Biostatistics, and Occupational Health

Education:

- PhD (Epidemiology), Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA 2008.
- MPH, Columbia University Mailman School of Public Health, New York, NY, USA 2004.
- BS, Biology (Major), Economics (Minor), The College of New Jersey, Trenton, NJ, USA 2002.

2 Personal statement

My research is predicated on the notion that, metaphorically speaking, we are subjects in a social science experiment, where the experimenters are government, non-governmental, and private industry “stakeholders” who are pulling levers, often without knowledge of their potential effects on human health and wellbeing. Research is needed to identify these policy experiments, as well as their impacts, and this evidence must be leveraged to support decision making. My research and training initiatives have been designed to advance our understanding of the macro-level determinants of population health and health inequality, including the effects of population-level programs and public policies, with the objective of identifying solutions for improving health and health equity.

As it relates to this proposal, the process governing the distribution of health research funding is, of course, the primary determinant of how limited resources are allocated to generate evidence that can be used to promote population health, both in Canada and internationally. This process should be fair, transparent, and equitable, and research (on research) is needed to examine the implications of funding mechanisms. My work in impact evaluation is directly relevant to the rigorous examination of the impacts of various funding procedures, such as the implications of grant review panel discussions.

Since my first academic appointment at McGill University in 2010, my research has been supported by salary support awards through the Canada Research Chairs Program (2011-2021), approximately \$6 million in research support as NPI or co-PI, and about \$4.7 million in research support as a co-investigator, and I have served as the co-PI on a \$300,000 training grant funded by Universities Canada (total funding=\$12 million). I have published more than 115 peer-reviewed papers, as well as several reports and book chapters.

3 Most significant contributions

Evaluating public policies and their influence on health and health inequality in a global context

Few studies have rigorously examined the health and social effects of various public policies, particularly those related to gender and poverty, although they are hypothesized to influence the social determinants of health. I have helped to establish a program of research that evaluates the effects of social policies on health and health inequality using observational, quasi-experimental, and experimental methods. I have worked with students and colleagues to develop innovative causal research methods to identify the impacts of health and social policies, and I co-designed and led an international training program in these methods.

Relevant papers:

- Oppong-Nkrumah O., Kaufman J.S., Heymann J., Frank J. & **Nandi A.** The impact of increasing the minimum legal age for work on school attendance in low- and middle-income countries. *SSM - Population Health* 2019, doi: <https://doi.org/10.1016/j.ssmph.2019.100426>.
- **Nandi A.**, Charters TJ, Quamruzzaman A, Strumpf EC, Kaufman JS, Heymann J, Mukherji A, Harper S. Health care services use, stillbirth, and neonatal and infant survival following implementation of the Maternal Health Voucher Scheme in Bangladesh: A difference-in-differences analysis of Bangladesh Demographic and Health Survey data, 2000-2016. *PLOS Medicine* 2022;19:e1004022. <https://doi.org/10.1371/journal.pmed.1004022>
- Charters TJ, Kaufman JS, **Nandi A.** A causal mediation analysis for investigating the effect of a randomized cash transfer program in Nicaragua. *American Journal of Epidemiology* 2023;192(1):111-121. doi: <https://doi.org/10.1093/aje/kwac165>.
- Ishola F, Rosario C, Griffin S, Khosa C, **Nandi A.** Abortion legal reform and neonatal mortality in Mozambique. *Maternal and Child Health Journal* 2024; <https://doi.org/10.1007/s10995-023-03876-1>.

Contributions to the literature on the impacts of paid leave on health and wellbeing

A major focus of my research has been investigating of the impact of paid leave policies. My work on this topic initially focused on the impact of maternity leave expansion in low and middle-income countries, but then expanded to parental and paid adult medical leave policies in Europe and OECD countries. This research has contributed to evidence on the protective impact of maternity leave for infants and has provided important tools for policymakers who wish to design effective paid leave policies. For example, work by our team, in collaboration with colleagues from the UCLA WORLD Policy Analysis Center, was used by a major initiative spearheaded by the Clinton and Bill and Melinda Gates Foundations, called No Ceilings, to advocate for social reforms that support gender equity worldwide.

Recent papers:

- **Nandi A.**, Hajizadeh M, Harper S, Koski A, Strumpf, E, Heymann J. “Increased duration of paid maternity leave lowers infant mortality in low- and middle-income countries.” *PLOS Medicine* 2016;13(3):e1001985.
- **Nandi A.**, Jahagirdar D, Dimitris MC, et al. The impact of parental and medical leave policies on socioeconomic and health outcomes in OECD countries: a systematic review of the empirical literature. *The Milbank Quarterly* September 2018; doi: 10.1111/1468-0009.12340.
- Chai Y, **Nandi A.**, Heymann J. Association of increased duration of legislated paid maternity leave with childhood diarrhea incidence in low- and middle-income countries: A difference-in-differences analysis. *Journal of Epidemiology and Community Health*. 2020; 74(5):437–444. doi: 10.1136/jech-2019-212127
- Jahagirdar D, Dimitris M, Strumpf E, Kaufman J, Harper S, Heymann J, Atabay E, Vincent I, **Nandi A.** Balancing work and care: the effect of paid adult medical leave policies on employment in Europe. *Journal of Social Policy* 2021;50:552-68. doi:10.1017/S0047279420000264

Social interventions for improving population health

I collaborate with nongovernmental organizations to examine the effects of social programs on health and health inequality using quasi-experimental and experimental methods. This includes: a cluster-randomized trial evaluating the impact of providing affordable daycare in rural Rajasthan, India, with the NGO Seva Mandir; a quasi-experimental study for measuring the effects of health systems strengthening interventions in the Tabora region of Tanzania with CARE Canada/Tanzania; and measuring the effect of a baby-friendly hospital program with the NGO FHI 360’s Alive & Thrive initiative in Southeast Asia.

Recent papers:

- **Nandi A**, Agarwal P, Chandrashekar A, Maloney S, Richardson R, Thakur L, Harper S. Access to affordable daycare and women's mental health in Rajasthan, India: Evidence from a cluster-randomized social intervention. *Journal of Global Health* 2024;14:04063.
- doi: <https://doi.org/10.7189/jogh.14.04063>.
- Richardson R, Harper S, Schmitz N, **Nandi A**. The effect of affordable daycare on women's mental health: evidence from a cluster randomized trial in rural India. *Social Science & Medicine* 2018;217:32-41.
- **Nandi A**, Agarwal P, Chandrashekar A, Harper S. Access to affordable daycare and women's economic opportunities: early evidence from a cluster-randomized intervention in India. *Journal of Development Effectiveness* 2020;12:219-39.
<https://doi.org/10.1080/19439342.2020.1773898>
- Joyce C, Hou SS-Y, Ta BTT, Vu DH, Mathisen R, Vincent I, Nguyen VD, **Nandi A**. The association between a novel baby-friendly hospital program and equitable support for breastfeeding in Vietnam. *International Journal of Environmental Research and Public Health* 2021;18:6706.
<https://doi.org/10.3390/ijerph18136706>
- Hetherington E, Harper S, Davidson R, Festo C, Lampkin N, Mtenga S, Teixeira C, Vincent I, **Nandi A**. The impact of health care provider and community health worker trainings on maternal health outcomes in Tanzania. *Journal of Epidemiology and Community Health* 2023;77(6):410-416. doi: 10.1136/jech-2022-219995.

Building open-access longitudinal policy databases

I have led a team of researchers investigating and coding social policies around the world to create longitudinal policy databases dating back to 1995. Ten databases of labor and child protection policies are freely available online and have been downloaded and used by other investigators for primary research initiatives, resulting in publications and the generation of policy-relevant evidence. Our team is working on a database of early childhood education policy in high-income countries in the areas of access, quality and governance. This policy data is a unique resource that, in combination with suitable outcomes data, allows for in-depth study of the relation between public policies and health in a way that would not otherwise be possible. We are encouraged by the use of our policy databases by international researchers to conduct policy analyses and produce primary research outputs, including several publications of the effectiveness of child marriage legislation.

Recent papers:

- Majid MF, Mendoza Rodríguez JM, Frank J, Harper S, **Nandi A**. Do minimum wages improve early life health? Evidence from developing countries. *Social Science & Medicine* 2016;158:105-13.
- Atabay E, Vincent I, Raub A, Heymann J, **Nandi A**. Data Resource Profile: PROSPERED Longitudinal Social Policy Databases. *International Journal of Epidemiology* 2019, 1743–1743g, <https://doi.org/10.1093/ije/dyz153>.
- Ponce N, Shimkhada R, Raub A, Daoud A, **Nandi A**, Richter L & Heymann J. The association of minimum wage change on child nutritional status in LMICs: A quasi-experimental multi-country study. *Global Public Health* Vol. 0, Iss. 0, 2017; doi: 10.1080/17441692.2017.1359327.
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4 Other contributions to knowledge

Contributions to policy formation

Our research outputs are being used to support evidence-based efforts for improving health. For example, a longitudinal policy database created under a CIHR Programmatic Grant was used by a major initiative spearheaded by the Clinton and the Bill and Melinda Gates Foundations, called No Ceilings, to advocate for social reforms that support gender equity worldwide. I have a long-standing collaboration with the non-profit FHI 360, which manages the Alive & Thrive initiative in Southeast Asia, which has resulted in several research policy-relevant research outputs. We also provide research evidence to support government and intergovernmental organizations; for example, our analyses of the impact of paid maternity leave was cited in the Congress of the Philippines to support the passage of legislation for expanding maternity leave benefits.

Invited talks (international, past 5 years)

- Evaluating the impact of access to universal early childhood education on adolescent health in European countries. Department of Public Health, University of Copenhagen. Copenhagen, DE. Mar 2025
- Quasi-experimental research designs in epidemiology: what, why, how? Society for Epidemiologic Research Digital (SERdigital). [Webinar]. Nov 2024
- Prioritizing child health and wellbeing in policymaking. UK Academy of Medical Sciences. London, UK. Feb 2024
- Evaluating the impact of access to universal early childhood education on adolescent mental health in European countries. Health Research & Practice Weekly Seminar; Milken Institute School of Public Health; The George Washington University. [Webinar]. Nov 2023
- The impact of community-based daycare on health and well-being in Rajasthan, India. NYU Grossman School of Medicine; Department of Population Health; Division of Epidemiology & Section for Global Health. [Webinar] Dec 2020

Ad hoc peer review

- Epidemiology: *American Journal of Epidemiology, Annals of Epidemiology, Epidemiology, International Journal of Epidemiology, Journal of Epidemiology and Community Health*
- Public health and medicine: *American Journal of Hematology, American Journal of Industrial Medicine, American Journal of Public Health, BMC Infectious Diseases, BMJ Open, Canadian Journal of Public Health, Drug and Alcohol Dependence, Drug and Alcohol Review, Global Public Health, Health & Place, JAMA, Journal of Health Care for the Poor and Underserved, Journal of Nervous and Mental Disease, Journal of Psychiatric Research, Journal of Traumatic Stress, Journal of Urban Health, PLOS Medicine, PLOS One, Preventive Medicine, Social Psychiatry and Psychiatric Epidemiology, Tobacco Control, WHO South-East Asia Journal of Public Health*
- Health and social policy; social sciences: *Health Affairs, Health Economics, Journal of Development Studies, Social Science & Medicine, Social Science & Medicine Population Health, Social Science Quarterly, World Development*

Grant review (panel member)

- CIHR Project Scheme; Public, Community & Population Health Committee; 2023, 2021, 2020, 2019, 2016
- CIHR Foundation Scheme Stage 1; Public, Community & Population Health Committee; 2018

Grant review (ad hoc): Swiss National Science Foundation, 2023; Canada Research Chair (Tier II), 2022; Canada Research Chairs (Tier II), 2018; Canada Research Chairs (Tier II), 2016

5 Supervisory and mentoring activities

International training

Few training programs targeted to social epidemiologists provide an adequate foundation for applying experimental or quasi-experimental methods for evaluating the effects of interventions. I aim to

strengthen our international training capacity and cultivate human capital in this area. As part of my CIHR Programmatic Grant in Health Equity (2011-2016), I worked with our network of co-investigators to develop an international training network focused on methods for Impact Evaluation, which I expanded through my CIHR Foundation Grant (2016-2021). Nearly 30 post-graduate trainees from McGill University or our partner sites, including the Indian Institute of Management in Bangalore, UCLA Fielding School of Public Health, and Cayetano Heredia University in Peru, have been supported through our training program. For many trainees, we identified a team that included at least one faculty mentor from McGill and another from an international partner. Fellows were invited to participate in an intensive 2-week program that included training in methods for impact evaluation and different professional development workshops, after which they worked with their faculty mentors to develop a research question, design a study, and undertake their analyses. Several trainees published their results, we had at least two cases when Master's students pursued PhD studies, and several of our doctoral fellows successfully transitioned to public sector or faculty positions, including positions at Cayetano Heredia University, McGill University, and the University of California Berkeley.

Mentorship

I have supervised five post-doctoral fellows (two co-supervised), 12 PhD students (four co-supervised), and 2 MSc students. This includes hosting one MSc trainee enrolled in the Erasmus Mundus Joint Master's Degree in Infectious Diseases & One Health. My trainees have been productive, having published nearly 50 first-authored papers in leading international journals. Many have successfully obtained funding from FRQS and CIHR, with one student receiving a prestigious Vanier Canada Graduate Scholarship. Several of my PhD students and post-doctoral fellows have transitioned to faculty jobs, including positions at Dalhousie University, Emory University, George Washington University, McGill University, the University of Heidelberg, the University of Toronto, and the University of Sherbrooke, whereas others have entered private industry or the public sector. I have also supervised many undergraduate and graduate interns and practicums.

- *Postdoctoral fellows*: José Ignacio Nazif-Muñoz (co-supervised with M Ruiz-Casares), 2016-2018; M. Farhan Majid (co-supervised with S Harper), 2014-2016; Mohammad Hajizadeh, 2012-2014; Juliet Rumball-Smith, 2012-2013; Drissa Sia, 2012-2015
- *PhD students*: Jasleen Arneja, 2021-present; Sarah Windle, 2020-present; Sherry Hou, 2019-present; Caroline Joyce, 2019-2024; Foluso Ishola, 2018-2022; Brice Batomen (co-supervised with L Moore), 2017-2020; Oduro Oppong-Nkrumah (co-supervised with J Kaufman), 2015-2020; Robin Richardson (co-supervised with S Harper), 2015-2018; Catherine Arsenault (co-supervised S Harper), 2013-2017; Lauren Maxwell, 2013-2017, Alissa Koski, 2012-2016; Erika Braithwaite, 2012-2016
- *Master's students*: Hector Gonzalez Dorta (Erasmus Mundus Joint Master's Degree in Infectious Diseases & One Health), 2021-2022; Catherine Verreault, 2016-2019

6 Other relevant information None.

The Influence of Reviewer Expertise and Engagement on Peer Review of Grants

Objectives

The overall aim of this project is to investigate how the peer review process may affect reliability and fairness in the distribution of scores and funding success of CIHR Project Grant applications.

Specific Aims

1. To evaluate how reviewer expertise and engagement affects the impact of grant panel discussion on application scores.
2. To assess whether the effects of reviewer expertise and engagement differ by applicant characteristics such as gender or career stage.
3. To evaluate potential alternative schemes such as differential weighting of scores or partial randomization to determine funding for qualified applications.

Background and Rationale

Peer Review Model for Grants. CIHR's Project Grant Program annually invests hundreds of millions of dollars in proposals to fund the "best ideas" and advance the health of Canadians (1). Prioritizing research through funding comes from both strategic investments and an evaluation of competitive applications by Peer Review Committees (PRCs). PRCs covering over 50 basic and applied research domains are composed of roughly 20-30 members identified and assigned based on having sufficient expertise and experience to evaluate, rate, and discuss applications (2).

Grant review panels using PRCs are common in many agencies and countries (3–5). The rationale for using panels is that committees encompass a diverse range of perspectives and overlapping expertise that can correct misunderstandings among panel members, clarify ambiguities in scoring justifications, and converge on a more calibrated assessment (3). However, committees may also suffer from a range of potential biases that affect group decision making, such as groupthink, conformity, free riding, polarization, or the influence of asymmetric information (5). Such biases may be exacerbated when reviewers and non-reviewers of applications have access to different information.

A key challenge in grant panel review is human resource limitations (3). PRCs receive many submissions, so each application is only evaluated by 3 reviewers who thoroughly read each application and identify its strengths and weaknesses with respect to the main adjudication criteria (significance, approaches and methods, expertise and resources). Based on the initial reviewer scores, the bottom-scoring 60% of applications are streamlined, with the remaining applications discussed by the panel. In the meeting the 3 reviewers of each application provide their initial scores (on a scale from 0 [poor] to 4.9 [outstanding]), and summarize the application's strengths and weaknesses, followed by panel discussion to arrive at a consensus score among the three reviewers. Each panel member then provides a final score (0 to 4.9) that must be within +/- 0.5 of the consensus score. The average of these scores, with equal weight given to all panel members, are used as inputs to the final decisions regarding funding.

The discretion of panel members to deviate from the consensus score could have meaningful consequences for the final score and ranking of applications (6–8). Panel members' scores may be influenced by reviewer characteristics such as expertise regarding the application, level of engagement (reviewer or non-reviewer), gender, and prior reviewing experience. Such influences may also vary by characteristics of the applicant (gender, career stage). These factors may be particularly influential for panel members who *do not review* the application and must rely on panel discussion to form their final scores. Thus, in panels where only a few members have read the

application, uneven access to information may lead to less-than-fair judgements (e.g., an influence of non-experts on final scores). The dynamics of group interactions, pressure to conform, and the subjective nature of discussions may all play critical roles in shaping which grants ultimately receive funding (3,5,7). These challenges may also be affected by virtual rather than face-to-face panel discussions (9,10).

Prior Literature. Most prior research on grant peer review has focused on reviewer reliability (4). Several studies suggest that inter-rater reliabilities for applications are moderate or low (5,11–16), vary across committee domains (11), and require many reviewers to be consistent (17,18). However, low reliability is not necessarily inconsistent with identifying quality applications and can reflect a healthy diversity of opinions, expertise, or value judgements about research quality (19,20). There are also longstanding concerns that grant peer review systematically disadvantages some groups, including by gender (11,21–23), race or ethnic background (12,24), and career stage (8,11). Panel discussions could lead to final scores that reduce the competitiveness of applications from under-represented groups if non-reviewing members rely on heuristics (such as applicant reputation or institution) rather than on a careful evaluation of content.

Differences between consensus scores and the final scores from the full panel may reflect systematic impacts of panel member expertise and engagement. However, evaluations of panel discussions in grant peer review are rare (4). Past work shows that the expertise, experience, and level of engagement (reviewer vs non-reviewer) of each panel member can affect the distribution of final scores (11,25–27). However, little evidence exists on whether these factors may also affect the difference between consensus and final scores within a committee, which has implications for fairness. Prior work suggests that committee discussions fail to improve consistency and often demonstrate “strategic rating” among panel members to influence funding of specific applications (5,16,25,28). Direct observation of committee discussions also suggests concerns about fairness due to the influence of committee chairs, variation in the length of discussion and discussion topics, and attempts to ensure the funding of specific applications (5,10,16,28). Overall, the prior literature raises concerns that the dynamics of group decision making in grant panels can impede efficiency and exacerbate inconsistencies in scoring rather than mitigate them (16,20).

Finally, prior evidence suggests that not only can panel review affect changes in the scoring of applications, it can have important consequences for which applications are ultimately funded, particularly those in the middle range of ‘potentially fundable’ applications. Prior work suggests that anywhere from 10% to 30% of application funding results are affected by discussion (10,13,26–29).

Evidence Gaps. Prior evidence suggests the need for a systematic evaluation of how reviewer expertise and engagement may affect grant panel outcomes. Furthermore, whether these effects may disproportionately impact applicants that have historically experienced barriers to funding is unknown—this has important implications for equity in grant funding. Finally, there remains a need to explore whether alternative schemes such as weighting votes by reviewer engagement status or partial randomization of funding (30) could produce more reliable outcomes.

Methods

Data and Design

This is an observational study and we will utilize restricted application-level data provided by agreement with CIHR’s Funding Analytics Team. The dataset will include application-level information on funding result and amount, initial reviewer scores, consensus score, keywords or domain of inquiry (e.g., biomedical, clinical, etc.), re-submission status, number of investigators, funding amount requested, and PRC. Key data on the panel members will include our main

covariates of interest: **self-described expertise** (high, medium, low, not enough expertise) and **engagement** (reviewer or non-reviewer), as well as gender, experience, career-stage, past funding success, and conflicts of interest. The Funding Analytics Team will also provide data on the gender and career stage of the applicants.

Statistical Plan

Aim 1. Our aim is to assess how changes from the initial consensus score to the final review scores may vary with panel expertise and engagement. Given the multilevel structure of the data (applications nested within committees and reviewers) we will use hierarchical models to account for clustering at the committee, reviewer, and application level). Our primary outcome will be the difference between consensus and final scores, which captures any impact of changes during panel discussion. A simple linear model for this data is:

$$d_{ijk} = \alpha + \beta_r Rev + \beta_e Exp + \gamma X + \delta Z + (u_{0j} + u_{0k} + \varepsilon_{ij})$$

Where d_{ijk} is the *difference* between the consensus and final score for the i th review of the j th application in PRC k . The key coefficients of interest are β_r and β_e that capture any differences in the change in scores by reviewer engagement or expertise. X and Z are vectors of applicant-level and application-level characteristics, respectively, and a compound error term in brackets () encompasses random effects at the reviewer and committee level plus a within-application independent Gaussian error term.

Aim 2. Given the existing evidence that the impact of reviewer expertise may vary with other characteristics (11,31), as well as evidence that applicant-level differences (e.g., gender, career stage) may also vary with other factors such as committee (22) we will expand the above model to assess whether reviewer engagement and expertise differentially affect applicants by gender and career stage.

Aim 3. Aim 3 will be exploratory and will evaluate the consequences of potential alternative evaluation schemes. For example, a modified scheme of upweighting scores by the level of engagement with the application or random assignment to funding of applications within a ‘grey-zone’ in the fundable range. We will compare the predicted distribution of outcomes by gender and career stage for these alternatives with those of the standard model above.

Feasibility

We have considerable past experience with multilevel data and evaluation (32–36). The data required for this project goes beyond publicly available data on the [Open Data Portal](#). We have confirmation from CIHR’s Funding Analytics Team that the application-level data required to assess the impact of reviewer characteristics on scores is feasible for extraction for analysis by our team. Data on applicant and reviewer personal characteristics (e.g., gender, career stage) is confidential and subject to additional security precautions. For these analyses we have agreed to supply computer code to the Funding Analytics Team, who will run the analyses and return the results to our research team.

Timeline

Personnel	Timeline of research activities	Year1	Year2	Year3
Harper, Nandi, Postdoc	Analysis plan, data, meeting with FA team			
Postdoc, RA	Lit review, OSF infrastructure, website			
Postdoc, Harper, Nandi	Analyses of Aims 1 & 2			
Postdoc, Harper, Nandi	Analyses of Aim 3			
Harper, Nandi, RA	Knowledge translation, dissemination			

Expected outcomes

The primary outputs from this project will be training in reproducible research practices (37), peer-reviewed publications, presentations, reproducible analytic code, and non-technical policy briefs. Our results can provide new insights to CIHR's ongoing discussions regarding how it structures and designs its peer review processes, as well as potential alternative reforms (e.g., 13,25). CIHR is also a core partner of the Research on Research Institute (RoRI), and the aims of this research bear directly on RoRI's project on peer review, including the recent RoRI [Atlas of Peer Review](#) project that aims to synthesize existing work on peer review and propose innovative methods for improving the quality of peer review across multiple domains, including funding (38). Our results will also be relevant to other competitions or agencies that use a similar multi-stage review process.

Training and mentoring

Our training objectives are to: (1) introduce trainees to rigorous methods for evaluating peer review; (2) strengthen trainees' data science skills via developing code and analysis plans to be executed remotely; (3) expose trainees to best practices for open, reproducible and ethical conduct of social sciences research; and (4) involve trainees in the development of a knowledge mobilization plan and strategies for communicating research findings to non-academic stakeholders. We will recruit and train one postdoctoral scholar for two years that will spearhead the project analysis, develop the code, and lead the writing of reports. We will also hire one part-time research assistant to systematically review all of the existing research on the impact of panel discussions on funding scores and outcomes, coordinate meetings and dialogue with the remote analysis team, create and maintain a project website, and set-up and be responsible for maintaining infrastructure for reproducibility via the Open Science Foundation.

Knowledge mobilization

Our primary KM plan is to publish and distribute our findings to stakeholders in the discussion around peer review reform, as well as public facing outputs in social media and a dedicated website. This includes CIHR and the RoR Institute. We aim to publish 3 papers in open access topical (e.g., Research Evaluation) and health (e.g., CMAJ, Lancet) journals and we will make all of our code and reports available on the Open Science Foundation repository. We plan to reach out to CIHR and the RoR Institute with our findings and to submit and present our work at the annual Metascience [conference](#), as well as the Canadian and US public health and policy conferences.

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Funds requested from SSHRC

The template below is used by different funding opportunities and includes up to 10 years of funding.

Ensure that your proposed budget does not exceed the maximum amount or the maximum number of years that may be requested as outlined in the instructions to applicants for the specific program or funding opportunity to which you are applying.

For each budget year, estimate as accurately as possible the research costs requested from SSHRC. For each Personnel costs category, enter the number of individuals to be hired and specify the total amount required. For each of the other categories, enter the total amount required.

Please ensure that annual totals correspond with those entered in the Proposed Budget section of your application.

	Year 1		Year 2		Year 3		Year 4		Year 5	
Personnel costs	No.	Amount	No.	Amount	No.	Amount	No.	Amount	No.	Amount
Student salaries and benefits/Stipends										
Undergraduate										
Masters	1	\$15,899	1	\$15,899	1	\$15,899				
Doctorate										
Non-student salaries and benefits/Stipends										
Postdoctoral	1	\$62,500		\$62,500						
Other										
Travel and subsistence costs										
Applicant/Team member(s)										
Canadian travel		\$1,410		\$2,500		\$3,910				
Foreign travel										
Students										
Canadian travel										
Foreign travel										
Other expenses										
Professional/Technical services										
Supplies										
Non-disposable equipment										
Computer hardware		\$4,000								
Other										
Other expenses (Specify)										
Open access publications (3x2500)				\$5,000		\$2,500				
Total		\$83,809		\$85,899		\$22,309		\$0		\$0

	Year 6		Year 7		Year 8		Year 9		Year 10	
Personnel costs	No.	Amount	No.	Amount	No.	Amount	No.	Amount	No.	Amount
Student salaries and benefits/Stipends										
Undergraduate										
Masters										
Doctorate										
Non-student salaries and benefits/Stipends										
Postdoctoral										
Other										
Travel and subsistence costs										
Applicant/Team member(s)										
Canadian travel										
Foreign travel										
Students										
Canadian travel										
Foreign travel										
Other expenses										
Professional/Technical services										
Supplies										
Non-disposable equipment										
Computer hardware										
Other										
Other expenses (Specify)										
Open access publications (3x2500)										
Total		\$0		\$0		\$0		\$0		\$0

TOTAL \$192,017

Budget Justification

Personnel costs (172697)

One postdoc for 2 years @\$50,000 per year salary plus 25% benefits (\$62,500). The rationale for a postdoc rather than a PhD student is the time constraints and level of skill needed to execute the work. This requires the development and application of a remote analysis plan for Aim 2, the application of multilevel models, methods for measuring bias and the need to engage with the literature on peer review in a short time frame. To help facilitate the above tasks we will hire one part-time research assistant over the course of the project @\$25/hr + 22.3% benefits x 10 hrs/wk x 156 weeks for a total of \$47,697. The main tasks of the RA will be to synthesize and systematically review all of the research on the impact of panel discussions, as well as to help coordinate meetings and dialogue with the CIHR Funding Analytics Team, and to set up and maintain the project's reproducible code infrastructure on the Open Science Foundation's website, and to design and maintain a website for the project itself that will contain and display research outputs.

Travel costs (7820)

Three team members (PI, co-PI, and postdoc) will make two trips to Ottawa (one early to discuss the datasets, analysis plan and mechanics of remote working, and one for the end of study as required by the RFP). We will need to travel to Ottawa in the first year to develop a detailed analysis plan with the Funding Analytics Team that can be executed remotely (For Aim 2), and to gain a better understanding of the details of the confidential fields of data on applicants that can be used to evaluate how reviewers may impact equity in grants. The second trip, as detailed in the RFP, will also be to Ottawa to participate in a knowledge mobilization event or conference to share findings of the research. For these trips we estimate \$720 in round-trip train travel from Montreal to Ottawa (\$120 per person x 3 people x 2 trips), \$1200 in hotel costs (\$200 per night for 1 night x 3 people x 2 trips) and per diem costs of \$450 (McGill local rate of \$75/day for 2 days x 3 people x 2 trips). We also budget \$2500 for 1 scientific conference presentation in Year 2 for the postdoc (we plan on the annual MetaScience conference, likely for 2026 or 2027) and \$2500 for 1 conference for the PI in Year 3 to present overall findings at a public health conference (Canadian Society of Epidemiology and Biostatistics or Society for Epidemiologic Research 2027).

Other expenses (11500)

Non-disposable equipment

The recruited postdoc and research assistant will each require a laptop for conducting relevant background research, coding policies, analyzing data, writing up manuscripts and theses (2 x \$2000 based on McGill University pricing). Total cost: \$4,000.

Publication fees

We plan to make our research results widely available and estimate publishing 3 manuscripts: 1) Aim 1: evaluating the impact of reviewer expertise and role on overall scores and funding; 2) Aim 2: variations in the impact of reviewer expertise and role on funding by gender and career stage; Aim 3: evaluation of potential alternative strategies for rating and funding applications. As per the KMB plan we have specific journals targeted for these manuscripts, and we will publish these in open access format at an estimated cost of \$2500 per manuscript. Total cost: \$7,500.